Rank-four Grothendieck rings through multiplicity 128

Abstract

We classify the complex-categorifiable rank-four fusion rings whose maximum structure constant is at most 128. Exactly fifteen of the 889 530 rings are categorifiable, and all fifteen have unitary realizations. The five existence classifications—general, pivotal, spherical, pseudo-unitary and unitary—therefore coincide in this range. The proof uses a fixed collection of arithmetic and central-induction obstructions. Its principal additional ingredient is a positive measure of squared central dimensions in a sphericalization, together with the central sign object and the first two Frobenius–Schur indicators. A parity constraint from skew self-duality supplies a further obstruction to sphericalization. All finite applications are generated and verified automatically by exact certificates. The checker exhausts every required signed table and induction branch; no exclusion is assigned by a catalogue label. The input is the rank-four census at a pinned repository commit. The exclusions use general categorical results, the obstructions proved here, and an automatic application of the published C_3 near-group classification. Unrestricted structural results are treated separately.

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1 The finite classification

A *Grothendieck ring* here means a fusion ring admitting a fusion-category categorification over $\mathbb{C}$. The question concerns existence for a based ring, not a classification of all tensor categories realizing it. For structure constants N_{ij}^k, put $\mu(R) = \max_{i,j,k} N_{ij}^k$. We use the pinned rank-four census [1] and write $F_{m,j}$ for its jth ring of exact multiplicity m. For compatibility with historical lists, the identifiers through $m = 32$ follow the mapping supplied with the certificates; above 32 they follow the pinned parameter-file order. The supplement records the source index, tensor digest, and any historical basis permutation. Identifiers name based rings but are never premises of an obstruction test.

Theorem 1.1 (Bounded classification)**.** *A rank-four fusion ring R with $\mu(R) \le 128$ admits a fusion categorification over $\mathbb{C}$ if and only if it appears in Table 1. Every ring in that table admits a unitary categorification. Consequently general, pivotal, spherical, pseudo-unitary and unitary categorifiability give the same fifteen based rings in this range.*

The forward implication is the exclusion theorem proved below; the reverse implication uses the established models in the table. The logical reduction is particularly useful: a unitary realization establishes all five existence statements, whereas a general obstruction excludes all five. There is no need for five independent classification algorithms.

The census contains 889 530 based rings through multiplicity 128. Table 2 groups the exact counts in four disjoint ranges; the supplement gives the count at every exact multiplicity and a split by duality type. The input is the rank-four data of the fusion-ring census [1] at commit `7f1b36e74df53a8a1cf8597db45d261e408d61b2`. That repository supplies exhaustive coverage and exact counts through $m = 128$; Appendix B records the tensor interface needed by the classification checker. Every identifier is accompanied by a full tensor digest, so no mathematical conclusion depends on catalogue order. Every input ring is covered by a realization or nonexistence certificate.

The mathematical tools are presented before their finite application. Known results are cited with their hypotheses; the new inequalities and reductions are proved. The electronic supplement records applications as coefficient, polynomial, prime, sign and bounded-enumeration certificates, rather than as hundreds of separate case proofs. The companion paper studies what can be proved without a multiplicity bound and is not used to establish Theorem 1.1.

Ring	A unitary realization	Source for existence
$F_{1,1}$	$\mathrm{Vec}_{C_2\times C_2}$	[10]
$F_{1,2}$	$\mathrm{Fib}\boxtimes\mathrm{Vec}_{C_2}$	[10]
$F_{1,3}$	$\mathrm{Rep}(D_{10})$	[10]
$F_{1,4}$	$\mathrm{PSU}(2)_6$	[10]
$F_{1,5}$	$\mathrm{Fib}\boxtimes\mathrm{Fib}$	[10]
$F_{1,6}$	$\mathrm{PSU}(2)_7$	[10]
$F_{1,7}$	Vec_{C_4}	[10]
$F_{1,8}$	$\mathrm{TY}(C_3)$	[10]
$F_{1,10}$	the pseudo-$\mathrm{PSU}(2)_6$ ring	[10]
$F_{2,2}$	the even part of $2D2$	[2, Theorem 3.1]
$F_{2,15}$	the Haagerup category $\mathcal{H}_1$	[6, Table 2]
$F_{2,17}$	$\mathrm{Rep}(A_4)$	standard representation category
$F_{3,19}$	$\mathrm{Ad}(E_8)$	[5]
$F_{3,24}$	C_3 near-group, parameter 3	[7, Theorem 9.1]
$F_{6,81}$	C_3 near-group, parameter 6	[7, Theorem 10.19]

Table 1: The fifteen positive rings. Here D_{10} has order ten; Fib is unitary Fibonacci, and $\mathrm{PSU}(2)_k$ denotes the even part of the standard unitary $\mathrm{SU}(2)_k$ category. The word "pseudo" in the name of $F_{1,10}$ does not weaken its unitary realization.

Exact multiplicity range	Fusion rings	Categorifiable rings
1–32	17 736	15
33–64	104 672	0
65–96	268 810	0
97–128	498 312	0
Total	889 530	15

Table 2: The complete census and its categorifiable subset.

2 Arithmetic and signed dimension characters

2.1 Conventions and standard obstructions

A fusion ring has a distinguished basis $b_0 = 1, b_1, b_2, b_3$, nonnegative integral structure constants, duality $i \mapsto i^*$, and Frobenius reciprocity, with $N_{ij}^0 = \delta_{i,j^*}$. We use the row convention $(N_i)_{jk} = N_{ij}^k$. Every rank-four fusion ring is commutative: its complexification is semisimple of dimension four and has the one-dimensional Frobenius–Perron representation; a noncommutative block would already have dimension at least four.

Write $d_i = \mathrm{FPdim}(b_i)$ and $D = \sum_i d_i^2$. The four characters χ have formal codegrees

$$f_\chi = \sum_i \chi(b_i)\chi(b_{i^*}) = \sum_i |\chi(b_i)|^2.$$

They are the eigenvalues, with multiplicity, of

$$B = \sum_i N_i N_{i^*} = \sum_i N_i N_i^{\mathsf{T}}. \tag{1}$$

The following necessary conditions will be used in their stated scope.

Proposition 2.1 (Established inputs)**.** *1. In any complex categorification every formal codegree is a d-number. For its monic minimal polynomial $t^e + a_1 t^{e-1} + \cdots + a_e$, this means $a_e^j \mid a_j^e$ for $1 \le j \le e$ [12].*

2. If $S \subseteq R$ is a fusion subring, $D/\operatorname{FPdim}(S)$ is an algebraic integer [3, Theorem 7.17.6].

3. Every character value of a complex-categorifiable fusion ring is cyclotomic [4, Theorem 8.51 and Corollary 8.53].

4. In a spherical categorification with dimension character χ, every $\chi(b_i)$ is nonzero, $\chi(b_i) = \chi(b_{i^*})$, and its global dimension $\Delta = f_\chi$ satisfies $\Delta/f \in \overline{\mathbb{Z}}$ for every formal codegree f. Its dimension field contains all formal codegrees [13, Theorem 2.13 and Corollaries 2.14–2.15].

5. A weakly integral fusion category is pseudo-unitary [4, Proposition 8.24]. A categorifiable fusion ring with precisely two distinct simple Frobenius–Perron dimensions has a pseudo-unitary Galois conjugate [14, Lemma 3.1].

The last assertion concerns a Galois conjugate; it does not claim pseudo-unitarity of every categorification of the ring.

Lemma 2.2 (Galois reduction)**.** *If a spherical dimension character belongs to the Galois orbit of the Frobenius–Perron character, a spherical categorification has a pseudo-unitary Galois conjugate with the same fusion rules.*

Proof. Extend the required number-field embedding to an automorphism of $\mathbb{C}$ and conjugate the finite associativity, rigidity and spherical structure data. The resulting simple dimensions are $d_i > 0$, and its global dimension is D. □

The low-rank exclusions needed in the finite application are proved in Section 5 using these general inputs and central induction. The near-group branch also uses the explicit hypothesis check for Larson's classification stated in Section 5.

2.2 Small arithmetic certificates

For a nonzero algebraic integer u, put $e_u = [\mathbb{Q}(u) : \mathbb{Q}]$ and $n_u = |N_{\mathbb{Q}(u)/\mathbb{Q}}(u)|$.

Lemma 2.3 (Normalized norm obstruction)**.** *If u/v is an algebraic integer, then*

$$\frac{v_p(n_u)}{e_u} \ge \frac{v_p(n_v)}{e_v} \qquad \text{for every rational prime } p. \tag{2}$$

Proof. Taking the norm in $L = \mathbb{Q}(u, v)$ shows that $n_u^{[L:\mathbb{Q}]/e_u}/n_v^{[L:\mathbb{Q}]/e_v}$ is an integer. Taking its p-valuation gives the assertion. □

Thus a failed divisibility needs only two irreducible polynomials and one prime. As a preliminary shortcut, a signed Casimir determinant not divisible by a prime required by every possible Δ excludes the whole table: if the minimal polynomial of Δ had norm divisible by that prime, the characteristic polynomial of the Casimir would have constant term divisible by it too.

Lemma 2.4 (Prime certificates for fields)**.** *Let $g \in \mathbb{Z}[t]$ be monic irreducible and $K = \mathbb{Q}(\alpha)$, $g(\alpha) = 0$.*

1. If K contains a root of a monic irreducible polynomial f, then $\deg f$ divides $\deg g$.

2. *Suppose g splits into distinct linear factors modulo a prime p. If K contains all roots of a monic integral polynomial f, then f splits into linear factors modulo p, with repetitions allowed.*

3. *If the squarefree reduction modulo p of a monic irreducible polynomial has irreducible factors of unequal degrees, its root field cannot be cyclotomic.*

Proof. The first assertion is the tower formula. In the second, p does not divide the discriminant of g, hence not the index of $\mathbb{Z}[\alpha]$ in the ring of integers of K. Every residue field above p is therefore $\mathbb{F}_p$. Reduce the linear factorization of f in the ring of integers of K at any such prime. For the third, Dedekind's factorization theorem identifies the factor degrees with the cycle lengths of a Galois permutation of the roots. A transitive abelian permutation group acts regularly after quotienting its kernel, so all cycles of each element have the same length. Every subfield of a cyclotomic extension is Galois and abelian. □

These are rejection certificates: failure to find a prime certificate proves neither a field inclusion nor cyclotomicity. Positive root lists in a number field are instead verified by multiplying their linear factors and recovering the complete rational polynomial.

2.3 Sphericalization and exhaustive signed tables

Proposition 2.5 (Signed reduction)**.** *If a rank-four fusion ring R admits a complex categorification, there is a commutative associative signed table T with the same unit and duality, Frobenius reciprocity, and*

$$|T_{ij}^k| \le N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2. \tag{3}$$

It has a character χ nonzero on every basis element, equal on dual basis elements, with totally real values. Put $\delta = \sum_i \chi(b_i)^2$. For every formal codegree f of either R or T, δ/f is an algebraic integer and $f \in \mathbb{Q}(\chi(b_i) : i = 0, 1, 2, 3)$. If T becomes R after a change of basis signs respecting duality, R admits a spherical categorification.

Proof. Take the sphericalization $\mathcal{S} \to \mathcal{C}$ [13, Section 5.1]. The two lifts of each simple differ by the central sign object ε, of dimension -1. Choose self-dual lifts over self-dual simples and dual lifts over a dual pair. The first choice is possible because the dual has the same nonzero spherical dimension, whereas the two lifts have opposite dimensions. The signed quotient $K_0(\mathcal{S})/(\varepsilon + 1)$ has coefficients given by the differences of the two lift multiplicities. This proves (3) and the unit and reciprocity assertions. Its regular representation is closed under transpose, so its complexified algebra is semisimple. The spherical dimension character is a one-dimensional representation. The same dimension-four argument as above consequently makes this quotient commutative, also in the non-self-dual case. The other quotient, $K_0(\mathcal{S})/(\varepsilon - 1)$, is R. Thus $K_0(\mathcal{S})$ is commutative as well.

The eight formal codegrees of $\mathcal{S}$ are twice the four codegrees of R together with twice the four codegrees of T: each character extends with the two lift values equal or opposite. Its global dimension is 2δ. Proposition 2.1(4) applied to $\mathcal{S}$ gives the norm and field assertions. If $T = R$ after choosing lift signs, one lift multiplicity in every product is zero. The chosen lifts close under tensor product and duality and give a spherical fusion subcategory equivalent to $\mathcal{C}$. □

The table search is finite by (3). For self-duality there are ten independent coefficients; for one nontrivial dual pair there are six. The exact matrices and associativity equations are displayed in the verification guide. Changing lift signs is a finite gauge action; normalizing independent odd coefficients to be positive does not omit an orbit. Every zero-denominator

branch in the elimination is retained. A successful signed certificate rejects *all* resulting tables, by norms, dimension fields, spherical reduction, or the signed moment obstruction of Section 6.

3 Central induction and two twist moments

Let $\mathcal{C}$ be pseudo-unitary, with its canonical spherical structure, and let $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ have right adjoint I. For simples X_i and A, write $M_{iA} = [F(A) : X_i] = [I(X_i) : A]$. The four simple summands of $I(1)$ occur once, have twist one and dimensions D/f_χ [13, Theorems 2.5 and 2.13]. Let L be their four restriction columns, including the central unit column $(1, 0, 0, 0)^{\mathsf{T}}$. Then

$$MM^{\mathsf{T}} = B, \qquad L\mathbf{1} = B_{\bullet 0}, \qquad B - LL^{\mathsf{T}} \text{ is a nonnegative integral Gram matrix.} \tag{4}$$

Indeed $FI(X) = \bigoplus_j X_j \otimes X \otimes X_j^*$, and adjunction gives the first identity. All remaining columns have first entry zero, which gives the other assertions. In particular the remainder is positive semidefinite and entrywise nonnegative. A vector in its kernel is orthogonal to every remaining restriction column.

3.1 Higher vanishing indicators

The following statement holds at arbitrary rank. For an object V, let $I_{\mathrm{low}}(V)$ be the sum, with its multiplicities in $I(V)$, of those simple constituents that occur in $I(1)$.

Theorem 3.1 (Higher-indicator bound)**.** *Let $\mathcal{C}$ be a pseudo-unitary fusion category over $\mathbb{C}$, with its canonical spherical structure, and let $m \geq 1$. If $\nu_1(V) = \cdots = \nu_m(V) = 0$, then*

$$(m+1)\,\mathrm{FPdim}(I_{\mathrm{low}}(V)) \leq D\,\mathrm{FPdim}(V).$$

More strongly, the same inequality holds with $I_{\mathrm{low}}(V)$ replaced by the full twist-one part of $I(V)$. Equality forces every other constituent to have a nontrivial $(m+1)$st root of unity as twist. In particular, if every simple constituent of V is non-self-dual, then $3\,\mathrm{FPdim}(I_{\mathrm{low}}(V)) \leq D\,\mathrm{FPdim}(V)$.

Proof. Write $w_A = [I(V) : A]\,\mathrm{FPdim}(A) \geq 0$ and $T = D\,\mathrm{FPdim}(V)$. The indicator identity [11, Theorem 4.1] is $\sum_A w_A \theta_A^k = D\nu_k(V)$. Negative moments vanish by complex conjugation. Expanding the nonnegative polynomial

$$|1 + z + \cdots + z^m|^2 = (m+1) + \sum_{k=1}^{m} (m+1-k)(z^k + z^{-k}) \qquad (|z| = 1)$$

gives $\sum_A w_A |1 + \theta_A + \cdots + \theta_A^m|^2 = (m+1)T$. Every twist-one constituent contributes $(m+1)^2 w_A$; discarding the remaining nonnegative terms proves the bound. Equality requires the polynomial to vanish at every remaining twist, proving its stated support. The last assertion follows from additivity and $\nu_1(X) = \nu_2(X) = 0$ for a non-self-dual simple X. □

For $m = 1$ this is the low-part estimate explicit in the proof of [13, Theorem 2.21]. Its higher-indicator proof is included here. The arbitrary-sector extension is recorded in Appendix A.

3.2 The moment inequality

For a nontrivial simple X, put $d = \mathrm{FPdim}(X)$ and $\epsilon = \nu_2(X)$. The indicator formula [11, Theorem 4.1] gives, with $w_A = [I(X) : A]\,\mathrm{FPdim}(A) \geq 0$,

$$\sum_A w_A = Dd, \qquad \sum_A w_A \theta_A = 0, \qquad \sum_A w_A \theta_A^2 = D\epsilon. \tag{5}$$

Here the twists are roots of unity, $\epsilon \in \{-1, 1\}$ for a self-dual simple and $\epsilon = 0$ otherwise.

Proposition 3.2 (Two-moment bound)**.** *If L_X is any known contribution to the twist-one mass of $I(X)$, then*

$$(3d + \epsilon)L_X \leq Dd(d + \epsilon).$$

Consequently the following bounds require no choice of second indicator:

$$(3d + 1)L_X \leq Dd(d + 1) \qquad (X \cong X^*), \tag{6}$$

$$3L_X \leq Dd \qquad (X \ncong X^*). \tag{7}$$

One may take $L_X = \mathrm{FPdim}(I_{\mathrm{low}}(X)) = (LL^{\mathsf{T}}d)_X$.

Proof. For $P(z) = d + (d + \epsilon)z + dz^2$, expand $|P(z)|^2$ on the unit circle and use (5). This gives

$$\sum_A w_A |P(\theta_A)|^2 = Dd(3d + \epsilon)(d + \epsilon).$$

The twist-one terms contribute at least $(3d + \epsilon)^2 L_X$. Since $3d + \epsilon > 0$, division gives the assertion. The function $e \mapsto (d + e)/(3d + e)$ is increasing on $[-1, 1]$; replace ϵ by one in the self-dual case. □

The familiar first-moment bound $2L_X \leq Dd$ and the weaker Fejér bound $9L_X \leq D(3d+2)$ follow as well. The checker uses (6)–(7); these suffice for all the low-induction inequalities needed here. Positivity is essential: these bounds are not applied to arbitrary spherical categories with signed dimensions.

3.3 A finite restriction test

Proposition 3.3 (Exhaustive low-induction certificate)**.** *Fix the positive dimension character d. For each formal codegree f, form the set of vectors $v \in \mathbb{Z}^4_{\geq 0}$ satisfying*

$$v_0 = 1, \qquad \sum_i v_i d_i = D/f, \qquad v_i \leq \min\{B_{0i}, \lfloor \sqrt{B_{ii}} \rfloor\} \quad (i > 0). \tag{8}$$

Reserve $(1, 0, 0, 0)^{\mathsf{T}}$ for the central unit. Enumerate all choices of the other three columns, with the codegree multiplicities retained, whose sum together with the unit column is $B_{\bullet 0}$. If every such choice fails the entrywise or positive-semidefinite condition in (4), or one of (6)–(7), then R admits no pseudo-unitary categorification.

Proof. A genuine restriction column satisfies its dimension equation, and $M_{0A} = 1$ on the four constituents of $I(1)$. The first row and the diagonal of (4) give the two bounds in (8). Hence the actual low part occurs in the finite list and satisfies every tested necessary condition. □

The computation is only linear algebra over a number field. Express the four dimensions and D/f in a rational power basis, row-reduce the dimension equation, and enumerate its bounded free integer coordinates. When the four dimensions are rationally independent, there is at most one possible column for each f; a negative or nonintegral coefficient is already an obstruction. When they are dependent, no uniqueness is presumed. Full induction matrices, modular data and associators are never enumerated.

3.4 The non-self-dual quadratic family

For an integer $m \geq 0$, let $S(m)$ be the commutative based ring with basis $\{1, g, x, y\}$, involution $g^* = g$, $x^* = y$, and multiplication

$$g^2 = 1, \qquad gx = y, \qquad gy = x, \qquad x^2 = y^2 = g + m(x + y), \qquad xy = 1 + m(x + y).$$

Theorem 3.4. *For every $m \geq 2$, the fusion ring $S(m)$ admits no fusion categorification over $\mathbb{C}$.*

Proof. Put $d = m + \sqrt{m^2 + 1}$. Thus $d^2 = 1 + 2md$, and the Frobenius–Perron dimensions are $(1, 1, d, d)$. By Schopieray's Galois-conjugacy theorem for two-dimension fusion rings [14, Lemma 3.1], any fusion categorification would have a pseudo-unitary Galois conjugate with the same fusion rules. It is consequently sufficient to exclude pseudo-unitary categorifications.

Suppose one exists, and use its canonical spherical structure. Its dimension is $D = 2 + 2d^2 = 4 + 4md$. The four characters of $S(m)$, displayed by columns, are

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ d & -d^{-1} & i & -i \\ d & -d^{-1} & -i & i \end{pmatrix}.$$

Their formal codegrees are D, $2 + 2d^{-2}$, 4, 4, respectively. The formal-codegree description of induction therefore gives

$$I(\mathbf{1}) \cong \mathbf{1} \oplus A \oplus B \oplus C, \qquad \mathrm{FPdim}(A) = d^2 = 1 + 2md, \quad \mathrm{FPdim}(B) = \mathrm{FPdim}(C) = 1 + md,$$

where the four central simples are pairwise nonisomorphic and occur with multiplicity one. Indeed, every irreducible representation of this commutative rank-four ring is one-dimensional. Furthermore,

$$FI(\mathbf{1}) \cong \bigoplus_{T \in \mathrm{Irr}\,\mathcal{C}} T \otimes T^* \cong 4\mathbf{1} \oplus 2mX \oplus 2mY,$$

where $X^* \cong Y$ correspond to x, y. By adjunction each of $F(A)$, $F(B)$, $F(C)$ contains $\mathbf{1}$ once, and the displayed decomposition shows that none contains g. Since d is irrational, comparison of dimensions now forces

$$[F(A) : X] + [F(A) : Y] = 2m, \qquad [F(B) : X] + [F(B) : Y] = [F(C) : X] + [F(C) : Y] = m.$$

These statements determine the sum of the two low-part dimensions without requiring the individual restriction multiplicities:

$$\begin{aligned} L &= \mathrm{FPdim}(I_{\mathrm{low}}(X)) + \mathrm{FPdim}(I_{\mathrm{low}}(Y)) \\ &= 2m(1 + 2md) + 2m(1 + md) = 4m + 6m^2 d. \end{aligned}$$

Both X and Y are non-self-dual. Applying (7) to each of them and adding gives $3L \leq 2dD$. But

$$\begin{aligned} 3L - 2dD &= 12m + 18m^2 d - 2d(4 + 4md) \\ &= 12m + 18m^2 d - 8d - 8m(1 + 2md) \\ &= 4m + (2m^2 - 8)d > 0 \end{aligned}$$

for $m \geq 2$, a contradiction. □

4 A quadratic root-of-unity obstruction

For $n \geq 1$, let R_n be the self-dual commutative rank-four based ring whose fusion matrices in the basis $(1, X, Y, Z)$ are

$$N_X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & n-1 & n-1 & n \\ 0 & n-1 & n & n \\ 0 & n & n & n \end{pmatrix}, \quad N_Y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & n-1 & n & n \\ 1 & n & n & n \\ 0 & n & n & n+1 \end{pmatrix}, \quad N_Z = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & n & n & n \\ 0 & n & n & n+1 \\ 1 & n & n+1 & n+1 \end{pmatrix}.$$

Put $\delta = (3n + \sqrt{9n^2+4})/2$. Direct substitution gives the dimension vector $(1, \delta - 1, \delta, \delta + 1)$ and

$$\delta^2 = 3n\delta + 1, \qquad D = 3 + 3\delta^2 = 6 + 9n\delta.$$

Lemma 4.1. *Let $d > 1$ be squarefree and let f be the conductor of $\mathbb{Q}(\sqrt{d})$: $f = d$ if $d \equiv 1 \pmod 4$ and $f = 4d$ otherwise. Put $\kappa = \sqrt{f/d}$ and $\lambda = (\varphi(f) - |\mu(f)|)/\kappa$. Here φ and μ denote Euler's totient and the Möbius function. If a sum of k roots of unity equals $a + b\sqrt{d}$, with $a, b \in \mathbb{Q}$, then $k \geq -a - \lambda b$.*

Proof. Let $K = \mathbb{Q}(\sqrt{d})$, let L be a common cyclotomic overfield, and write $T = \mathrm{Tr}_{L/K}/[L : K]$. Define $\ell(a + b\sqrt{d}) = -a - \lambda b$. We prove $\ell(T(\xi)) \leq 1$ for every root of unity ξ.

If m is the exact order of ξ and $f \nmid m$, then $K \cap \mathbb{Q}(\xi) = \mathbb{Q}$, because the conductor of K is f. Consequently $T(\xi) = \mu(m)/\varphi(m)$, whose absolute value is at most one. If $m = fh$, let χ be the primitive quadratic character modulo f and write $\xi = \zeta_m^r$, with $(r, m) = 1$. The induced quadratic Gauss sum gives

$$T(\xi) = \begin{cases} \dfrac{\mu(f)\mu(h) + \chi(r)\mu(h)\chi(h)\kappa\sqrt{d}}{\varphi(f)\varphi(h)}, & (f, h) = 1, \\ 0, & (f, h) > 1. \end{cases}$$

For clarity, this follows by splitting the units modulo m into the two values of χ: their sum over $\chi = 1$ is half the sum of the Ramanujan sum $\mu(m)$ and the induced Gauss sum. The latter equals $\chi(r)\mu(h)\chi(h)\sqrt{f}$ when $(f, h) = 1$ and vanishes otherwise; an additional power of a prime dividing f cancels upon summing over its last prime-power layer. The same formula includes nonsquarefree h through $\mu(h) = 0$. In the nonzero case, the definition of λ gives

$$|\ell(T(\xi))| \leq \frac{|\mu(f)| + \lambda\kappa}{\varphi(f)\varphi(h)} \leq 1.$$

Applying $\ell \circ T$ to the sum proves the lemma. □

4.1 The sharp conductor-forty boundary

Lemma 4.2 (A root-of-unity length certificate)**.** *Suppose that $\xi_1, \ldots, \xi_n$ are roots of unity and*

$$\sum_{j=1}^{n} \xi_j = a + b\sqrt{10}, \qquad a, b \in \mathbb{Q}.$$

Then $n \geq -a - 8b$. If equality holds, every ξ_j is either -1 or is a primitive fortieth root ζ_{40}^r with $\chi_{40}(r) = -1$, where $\chi_{40}(r) = (40/r)$ is the primitive quadratic Dirichlet character of conductor 40. Consequently, if

$$n = 18, \qquad \sum_{j=1}^{18} \xi_j = -2 - 2\sqrt{10},$$

then exactly two of the roots are -1, and the normalized absolute trace of $\sum_j \xi_j^2$ is 2.

Proof. Put $s = \sqrt{10}$. Let $K = \mathbb{Q}(s)$, whose conductor is 40, and choose a cyclotomic field L containing K and all the roots under consideration. Write

$$T(u) = \frac{\mathrm{Tr}_{L/K}(u)}{[L : K]}, \qquad \ell(a + bs) = -a - 8b.$$

We claim that $\ell(T(\xi)) \le 1$ for every root of unity ξ, with precisely the equality cases stated above. The normalized relative trace does not depend on the choice of common cyclotomic overfield.

Let m be the exact order of ξ. If $40 \nmid m$, the intersection $K \cap \mathbb{Q}(\xi)$ is $\mathbb{Q}$: otherwise the conductor of K would divide m. Restriction of automorphisms consequently gives

$$T(\xi) = \frac{\mu(m)}{\varphi(m)}.$$

It follows that $\ell(T(\xi)) \le 1$, with equality only when $m = 2$.

Now suppose $m = 40h$, and write $\xi = \zeta_m^r$, $(r, m) = 1$. Since $\mu(m) = 0$, the rational coefficient of $T(\xi)$ is zero. The quadratic coefficient is computed by the induced quadratic Gauss sum:

$$G_m(r) = \sum_{v \in (\mathbb{Z}/m\mathbb{Z})^\times} \chi_{40}(v)\zeta_m^{rv}.$$

For completeness, the standard induced-Gauss-sum identity gives

$$G_m(r) = \begin{cases} \chi_{40}(r)\mu(h)\chi_{40}(h)\, 2s, & (h, 40) = 1, \\ 0, & (h, 40) > 1. \end{cases}$$

Here $\mu(h) = 0$ already covers nonsquarefree h. This identity can also be obtained directly from the Chinese remainder theorem when $(h, 40) = 1$: the sum factors into the primitive Gauss sum $G_{40}(1) = \sqrt{40} = 2s$ and the Ramanujan sum $\mu(h)$, with the displayed quadratic character factors. If an extra power of 2 or 5 occurs in m, summing over the last such prime-power layer gives zero. The terms with $\chi_{40}(v) = 1$ are the automorphisms fixing K; their sum is $(\mu(m) + G_m(r))/2$. Since there are $\varphi(m)/2$ such automorphisms,

$$T(\xi) = \frac{G_m(r)}{\varphi(m)} = \frac{\chi_{40}(r)\mu(h)\chi_{40}(h)}{8\varphi(h)}s \quad \text{if } (h, 40) = 1,$$

and it is zero otherwise. Thus $|\ell(T(\xi))| \le 1$. Equality with value 1 forces $\varphi(h) = 1$ and $(h, 40) = 1$, hence $h = 1$, and then $\chi_{40}(r) = -1$. This proves the claim. Applying $\ell \circ T$ to the sum of the roots proves the inequality and its equality statement.

In the particular equality case, primitive fortieth roots have normalized absolute trace $\mu(40)/\varphi(40) = 0$, whereas -1 has trace -1. The absolute trace of $-2 - 2s$ is -2, so exactly two roots are -1. The square of a primitive fortieth root is primitive of order 20 and has normalized absolute trace $\mu(20)/\varphi(20) = 0$. The two other squares are 1, proving the last assertion. □

Proposition 4.3. *Let $n \ge 2$, write $9n^2 + 4 = t^2 d$ with d squarefree, and define λ as in Lemma 4.1. If $n = 2$, or if*

$$n^2 + nt\lambda > 8n^2 + 4, \tag{9}$$

then R_n admits no spherical categorification over $\mathbb{C}$. In particular this excludes every R_n with $2 \le n \le 127$, which is exactly the part of this family of multiplicity at most 128 apart from R_1.

Proof. We first derive a necessary root-of-unity identity for a pseudo-unitary categorification of R_n. Give such a category its canonical spherical structure, and use F, I for forgetful functor and central induction. The four characters, displayed by rows, are

$$\begin{pmatrix} 1 & \delta-1 & \delta & \delta+1 \\ 1 & \delta'-1 & \delta' & \delta'+1 \\ 1 & 0 & 1 & -1 \\ 1 & 1 & -1 & 0 \end{pmatrix}, \qquad \delta' = -\delta^{-1}.$$

The formal codegrees are $D, D' = 3 + 3(\delta')^2, 3, 3$. Accordingly,

$$I(1) = 1 \oplus A \oplus B \oplus C, \qquad d_A = D/D' = 1 + 3n\delta, \quad d_B = d_C = D/3 = 2 + 3n\delta.$$

Every summand has multiplicity one and twist one. Comparing the coefficients of 1 and δ in their restriction dimensions shows that their columns are

$$(1, a, 3n - 2a, a), \quad (1, b, 3n - 1 - 2b, b + 1), \quad (1, c, 3n - 1 - 2c, c + 1).$$

The first row of $MM^{\mathsf{T}} = Q := \sum_U N_U^2$, for the full induction matrix M, gives $a+b+c = 3n-1$.

Here the low part is forced without searching over a, b, c. Put $v = (0, 1, -2, 1)^{\mathsf{T}}$. Multiplication of the displayed matrices gives $v^{\mathsf{T}}Qv = 18$. On the three low columns, the values of v are

$$6(a - n), \quad 6(b - n) + 3, \quad 6(c - n) + 3.$$

Their squares sum to at most 18. The last two are odd multiples of three, so their squares already sum to at least 18. It follows that $a = n$ and $\{b, c\} = \{n - 1, n\}$. Thus

$$F(A) = (1, n, n, n), \quad F(B) = (1, n - 1, n + 1, n), \quad F(C) = (1, n, n - 1, n + 1).$$

Equality in the sum of squares also implies that v vanishes on every remaining column. Such a column is therefore

$$(0, b - t, b, b + t), \qquad b \geq |t|, \quad b > 0,$$

and its central dimension is $3b\delta + 2t$. Subtracting the four low columns from Q and comparing its entries gives

$$\sum b^2 = 6n^2 + 2, \qquad \sum bt = 2n, \qquad \sum t^2 = 2. \tag{10}$$

Thus precisely two high columns have nonzero t, which equals 1 or -1. Set $T = 3n\delta + 2$. Subtracting the first twist-trace equation for X from that for Z, then dividing by two, gives

$$\sum t(3b\delta + 2t)\theta = -T. \tag{11}$$

If the two signs of t differ, their b values differ by $2n$, by (10). Their positive dimensions then differ by $6n\delta + 4 = 2T$, contradicting the reverse triangle inequality in (11). Both signs cannot be negative. Hence both are positive; write their b values as p, q, with $p + q = 2n$. Their dimensions $\alpha = 3p\delta + 2$, $\beta = 3q\delta + 2$ satisfy

$$\alpha + \beta = 2T, \qquad \alpha\beta = (9pq + 4)\delta^2, \qquad T^2 = (9n^2 + 4)\delta^2.$$

If their twists are ζ, η, equation (11) implies

$$\zeta\eta^{-1} + \eta\zeta^{-1} = 2 - \frac{3(9n^2 + 4)}{9pq + 4}.$$

This rational number is an algebraic integer and lies in $[-2, 2]$. Since $pq \leq n^2$, it is at most -1. The value -2 would require $36pq = 27n^2 - 4$, which is impossible modulo nine. The

value -1 forces $pq = n^2$, hence $p = q = n$. The two twists then sum to -1 and are the primitive cube roots of unity.

All other columns have $t = 0$ and are of the form $(0, b, b, b)$. Their squared multiplicities sum to $4n^2 + 2$. Write $S = \sum b^2\theta$ over these uniform columns; it is a sum of $4n^2 + 2$ roots of unity, with repetitions. The first twist-trace equation for X gives

$$S = n^2 - n\delta.$$

For completeness, the low contribution is $n(1 + 3n\delta) + (2n - 1)T$; the two exceptional high columns contribute $-(n - 1)T$, so the equation is $0 = n(1 + 3n\delta) + nT + 3\delta S$, which is equivalent to the claimed identity using $\delta^{-1} = \delta - 3n$.

Writing $9n^2 + 4 = t^2 d$ with d squarefree, we have

$$S = -\frac{n^2}{2} - \frac{nt}{2}\sqrt{d}.$$

Lemma 4.1 therefore requires

$$4n^2 + 2 \geq \frac{n^2 + nt\lambda}{2}.$$

Condition (9) contradicts this inequality. For the finite parameters $3 \leq n \leq 127$, the supplement verifies it exactly from the squarefree part d of $9n^2 + 4$, its conductor and λ. These are scalar integer and rational calculations, not a search for roots of unity. The uniform estimate valid for every $n > 2$ is proved in the companion paper; it is not a premise of the finite certificate.

For $n = 2$, the same reduction gives eighteen roots whose sum is $S_1 = -2 - 2\sqrt{10}$. The second indicator of X is $\epsilon \in \{1, -1\}$. The same twist-trace equation with twists squared, using that the two exceptional twists still sum to -1, gives

$$S_2 = S_1 + \epsilon\frac{D}{3\delta} = -2 + (2\epsilon - 2)\sqrt{10}.$$

For $\epsilon = -1$, Lemma 4.2 applied to the eighteen squared roots requires at least 34 roots. For $\epsilon = 1$, the same lemma applied to S_1 shows that the normalized absolute trace of S_2 is 2, whereas $S_2 = -2$. Both cases are impossible.

Finally, a spherical categorification has a dimension character with no zero entries, so the displayed character table restricts it to the two quadratic characters. These are Galois conjugate; conjugating its structure constants if necessary would give a pseudo-unitary categorification. This completes the proof. □

5 Uniform applications of the moment bounds

The following consequences replace separate low-rank exclusion theorems. Throughout, the two-dimension Galois reduction is used only in the form stated in Proposition 2.1(5).

If g is invertible of order two, its second indicator is $\epsilon \in \{-1, 1\}$. Since $\mathrm{FPdim}(g) = 1$, equality in the triangle inequality for the second indicator forces $\theta_A^2 = \epsilon$ for every constituent A of $I(g)$. More generally, if H_X is any known mass in $I(X)$ supported on squared twist one, taking real parts of the second moment gives

$$2H_X \leq D(d_X + \nu_2(X)) \leq D(d_X + 1). \tag{12}$$

Indeed the remaining real trace is at least the negative of its mass. For a non-self-dual simple, equality in $3W_X \leq Dd_X$ forces the complementary twists to be primitive cube roots by Theorem 3.1.

5.1 Rank two from the first moment

Proposition 5.1. *If a fusion ring contains a based fusion subring with basis $1, x$ and $x^2 = 1 + ax$, where $a \geq 2$, it has no complex fusion categorification. This conclusion follows from the low-part moment bound, without invoking the rank-two classification.*

Proof. A categorification would give a fusion subcategory with this based ring. Its nontrivial Frobenius–Perron dimension d satisfies $d^2 = 1 + ad$ and $d > a$. The two-dimension reduction permits us to assume pseudo-unitarity of this subcategory. Its formal codegrees are $D = 1+d^2$ and $D' = 1 + d^{-2}$, and $D/D' = d^2$. Consequently

$$I(1) = 1 \oplus A, \qquad \dim A = d^2, \qquad F(A) = 1 \oplus aX.$$

The first-moment bound on $I(X)$ gives

$$2ad^2 \leq Dd.$$

Dividing by d and using $d^2 = 1 + ad$ gives $ad \leq 2$. But $ad > a^2 \geq 4$, a contradiction. □

The criterion applies whenever such a rank-two based subring is present.

5.2 A uniform diagonal two-invertible obstruction

Theorem 5.2. *Let R_m have self-dual basis $1, g, X, Y$ and products*

$$g^2 = 1, \quad gX = Y, \quad gY = X, \qquad X^2 = Y^2 = 1 + m(X + Y), \quad XY = g + m(X + Y).$$

For every integer $m \geq 3$, the ring R_m has no fusion categorification over $\mathbb{C}$.

Proof. Its positive dimensions are $(1, 1, d, d)$, where $d = m + \sqrt{m^2 + 1}$ and $d^2 = 1 + 2md$. The two-dimension reduction allows a pseudo-unitary assumption. Put $D = 4 + 4md$ and $V = X \oplus Y$.

The four dimensions in $I(1)$ are $1, d^2, 1 + md, 1 + md$. Moreover

$$FI(1) = 4 \cdot 1 + 2mX + 2mY.$$

In particular the low part of $I(g)$ is empty. Dimension comparison shows that the total multiplicities of X, Y in the last three restrictions are $2m, m, m$. Therefore their known twist-one mass in $I(V)$ is, independently of its allocation between X and Y,

$$W = 2md^2 + 2m(1 + md) = 4m + 6m^2d. \tag{13}$$

The total mass in $I(V)$ is

$$M = 2Dd = 8m + (16m^2 + 8)d.$$

The Gram entry $B_{gg} = 4$ implies that $I(g)$ is either twice one simple or a sum of four distinct simples. The first possibility contradicts its zero first twist trace. Write the four simple dimensions as $\beta_j = 1 + k_j d$: each restriction contains g once, no unit, and k_j copies of X, Y in total. Thus $k_j \in \mathbb{Z}_{\geq 0}$ and $\sum_j k_j = 4m$.

Put $\epsilon = \nu_2(g)$. All four twists have square ϵ. The zero first trace divides these four objects into the two possible twists, with mass $D/2$ at each. Since d is irrational, each side has two objects and the two k_j sum to $2m$. Each side therefore contributes at least

$$2m + 2m^2d$$

to $I(V)$, by $k_1^2 + k_2^2 \geq (2m)^2/2$.

If $\epsilon = 1$, the new twist-one mass is at least $2m + 2m^2 d$, disjoint from the low part. Adding (6) for X, Y yields

$$(3d+1)(6m + 8m^2 d) \leq 2Dd(d+1).$$

After reduction by $d^2 = 1 + 2md$, the left side minus the right side is

$$2(4m^2 - m - 4) + 2(8m^3 - 4m^2 - 3m - 4)d > 0$$

for $m \geq 3$. For example, substituting $m = n + 3$ gives coefficients $8n^2 + 46n + 58$ and $16n^3 + 136n^2 + 378n + 334$, respectively.

Suppose instead $\epsilon = -1$. The four new twists are $\pm i$, and their total mass in $I(V)$ is at least

$$H = 4m + 4m^2 d.$$

Remove this mass and the low mass W from the positive measure of $I(V)$. For the remaining measure, the first cosine moment is $-W$, its total mass is at most

$$M - W - H = (6m^2 + 8)d,$$

and its second cosine moment is at most

$$\frac{M + D(\nu_2(X) + \nu_2(Y))}{2} - W \leq \frac{M}{2} + D - W = 4 + (2m^2 + 4m + 4)d.$$

Cauchy–Schwarz therefore requires

$$W^2 \leq (6m^2 + 8)d(4 + (2m^2 + 4m + 4)d).$$

The difference between the two sides, reduced by the dimension equation, is

$$\begin{aligned} &8(m-2)(3m^3 + 3m^2 + 3m + 2) \\ &\quad + 8(m-2)(6m^4 + 6m^3 + 8m^2 + 5m + 2)d > 0. \end{aligned}$$

This excludes the second indicator choice as well. □

5.3 Propagation from an invertible involution

Proposition 5.3 (Residual-involution criterion)**.** *Let $\mathcal{C}$ be pseudo-unitary with commutative Grothendieck ring, and let $g \neq 1$ be an invertible simple with $g^2 = 1$. For its actual low restriction matrix L, put*

$$Q = B - LL^{\mathsf{T}}, \qquad W_i = (LL^{\mathsf{T}} d)_i.$$

Suppose $Q_{gg} = 2$ and $0 < W_g < D/2$. The two constituents of $I(g)$ outside $I(1)$ then have dimensions and twists

$$(D/2 - W_g, 1), \qquad (D/2, -1).$$

If moreover $D/2 - W_g = 1$, their restriction columns are respectively e_g and $Q_{g\bullet}^{\mathsf{T}} - e_g$. Consequently, for every nonunit simple $X_i \neq g$,

$$2W_i + DQ_{gi} \leq D(d_i + \nu_2(X_i)) \leq D(d_i + 1). \tag{14}$$

Proof. The residual Gram entry $Q_{gg} = 2$ is a sum of squares of nonnegative integer multiplicities. Thus exactly two additional simples occur, each once. Equality in the triangle inequality for the second indicator of g forces all twists in $I(g)$ to have square $\nu_2(g)$. The positive low mass has twist one, so $\nu_2(g) = 1$. The vanishing first trace divides the total dimension D equally between twists 1 and -1. Since $0 < W_g < D/2$, both additional sectors have positive mass and each contains one of the two remaining simples. Their dimensions follow.

When the first dimension is one, its restriction contains g and has Frobenius–Perron dimension one; hence its restriction is exactly g. Write v for the other restriction column. In the residual Gram product, only these two columns contribute to row g, so $Q_{gi} = \delta_{gi} + v_i$ for every i. For $i \neq g$, the low constituents and the twist-minus-one constituent therefore give disjoint squared-twist-one mass $W_i + (D/2)Q_{gi}$ in $I(X_i)$. Equation (12) proves the required bound. □

The criterion is finite and uniform: enumerate the admissible low parts by Proposition 3.3 and check (14) whenever its hypotheses hold. Uniqueness of an admissible low part is sufficient but is not assumed.

Corollary 5.4. *The ring with self-dual basis $1, g, r, x$ and products*

$$g^2 = 1, \quad gr = r, \quad gx = x, \quad r^2 = 1 + g + r, \quad rx = 2x, \quad x^2 = 1 + g + 2r + 5x$$

has no complex fusion categorification.

Proof. The ring is integral, so any categorification would be pseudo-unitary. The exact application verifier constructs its dimensions, formal codegrees and all admissible low parts from the multiplication tensor. It verifies that, up to permuting the low columns, there is a unique admissible low part and that this low part violates (14). This is the certificate recorded for $F_{5,33}$; no restriction column or numerical inequality is assumed as an input. □

5.4 Near-group root-length exclusions

Proposition 5.5. *Consider the near-group ring with invertible group $C_3 = \{1, g, g^2\}$ and one further basis object X, with*

$$gX = X, \qquad X^2 = 1 + g + g^2 + 3kX,$$

where $k \geq 1$ is an integer. Write

$$9k^2 + 12 = u^2 q, \quad q > 1 \text{ squarefree}, \qquad f = \begin{cases} q, & q \equiv 1 \pmod 4, \\ 4q, & q \not\equiv 1 \pmod 4, \end{cases} \qquad \lambda = \frac{\varphi(f) - |\mu(f)|}{\sqrt{f/q}}.$$

A necessary condition for a complex categorification is

$$ku\lambda \leq 7k^2 + 12. \tag{15}$$

Every parameter violating this inequality is excluded without invoking the near-group classification.

Proof. The two-dimension reduction again permits pseudo-unitarity. Put

$$d = \frac{3k + \sqrt{9k^2 + 12}}{2}, \qquad d^2 = 3 + 3kd, \qquad D = 6 + 3kd.$$

The number d is irrational: for $k \geq 2$, $9k^2 + 12$ lies strictly between $(3k)^2$ and $(3k + 1)^2$, and the case $k = 1$ gives 21.

The formal codegrees are $D, D', 3, 3$, where $d' = -3/d$ and $D' = 3 + (d')^2$. Their corresponding low central dimensions are $1, 1 + kd, 2 + kd, 2 + kd$. Since

$$FI(1) = 4 \cdot 1 + g + g^2 + 3kX,$$

comparison of rational and irrational coefficients forces, in row order $(1, g, g^2, X)$,

$$L = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & k & k & k \end{pmatrix}, \qquad B - LL^{\mathsf{T}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 3 & 1 & 2k \\ 0 & 1 & 3 & 2k \\ 0 & 2k & 2k & 6k^2 + 6 \end{pmatrix}.$$

The low twist-one mass in each of $I(g), I(g^2)$ equals $2 + kd = D/3$. Since these objects are non-self-dual, the two-moment bound is saturated. Every remaining twist in each induced object is consequently a primitive cube root of unity. Write these roots as ω, ω^2; the zero first trace forces mass $D/3$ at each.

Each of $I(g), I(g^2)$ has three further constituents, with multiplicity one. Their common constituents number exactly one, because the residual (g, g^2) Gram entry is one. The common constituent has dimension $2 + hd$ for an integer $h \geq 0$; every noncommon constituent has dimension $1 + pd$ for an integer $p \geq 0$.

For either induced object, comparing rational coefficients in its two nontrivial twist sectors shows that the common constituent lies alone in one sector, and the two noncommon constituents form the other. Thus $h = k$. Exchange ω, ω^2 if necessary so that the common constituent has twist ω. Write the four noncommon X multiplicities as p_1, p_2, p_3, p_4, where

$$p_1 + p_2 = k, \qquad p_3 + p_4 = k.$$

They all have twist ω^2. Put

$$S = \sum_{j=1}^{4} p_j^2 \geq k^2, \qquad t = S - k^2 \in \mathbb{Z}_{\geq 0}.$$

Every further central constituent restricts to nX for some positive integer n. The residual (X, X) Gram entry gives

$$\sum n^2 = 6k^2 + 6 - k^2 - S = 4k^2 + 6 - t.$$

Let $\Theta = \sum n^2 \theta$, with each twist repeated n^2 times. The first twist trace for $I(X)$ is

$$0 = (5k + 3k^2 d) + (2k + k^2 d)\omega + (2k + Sd)\omega^2 + d\Theta.$$

Using $3/d = d - 3k$ and $1 + \omega + \omega^2 = 0$ gives

$$\Theta = k^2 - kd - t\omega^2.$$

Consequently $\Theta + t\omega^2$ is a sum of exactly $4k^2 + 6$ roots of unity equal to

$$k^2 - kd = -\frac{k^2}{2} - \frac{ku}{2}\sqrt{q}.$$

The quadratic root-of-unity bound, Lemma 4.1, now gives

$$4k^2 + 6 \geq \frac{k^2 + ku\lambda}{2},$$

which is (15). □

The application verifier recognizes this entire family from its tensor, computes the squarefree part and the conductor, and checks the strict failure of (15) in exact arithmetic. Its applications are recorded in the generated certificate ledger. Only the first two indicator identities are used; the inequality does not by itself classify near-group rings at every parameter.

5.5 Automatic use of the near-group classification

Proposition 5.6 (A cited exclusion with checked hypotheses)**.** *A C_3 near-group ring*

$$g^3 = 1, \qquad gX = Xg = X, \qquad X^2 = 1 + g + g^2 + mX$$

can admit a complex categorification only if $m \in \{0, 2, 3, 6\}$.

Proof. It has exactly two distinct simple Frobenius–Perron dimensions, so any categorification has a pseudo-unitary Galois conjugate by Proposition 2.1(5). The necessary list then follows from [8, Theorem 1.1]. □

The executable test compares the entire multiplication tensor after each permutation of the nonunit basis, verifies the two-dimension hypothesis, and checks the parameter against the stated set. It applies at every multiplicity. The root-length calculation above remains useful as an independent arithmetic application, but no separate proof of Larson's classification is needed for this branch.

6 Positive central moments and the sign object

We use the canonical spherical structure on the sphericalization $\mathcal{S}$ of a hypothetical fusion category. Its realization as an equivariantization for the canonical involutive pivotal functor gives a central copy of $\mathrm{Rep}(C_2)$; see [16, Section 5.3]. Its central sign object ε has order two, dimension -1, and self-braiding $+1$. Consequently its twist in $\mathcal{Z}(\mathcal{S})$ is -1. The two lifts of each original self-dual simple are self-dual; their dimensions and second Frobenius–Schur indicators have opposite signs. The latter assertion follows directly from the definition of the second indicator: the two lifts have the same evaluation and coevaluation maps after forgetting equivariance, while their pivotal maps differ by -1.

Lemma 6.1 (A positive measure without pseudo-unitarity)**.** *Let $\mathcal{S}$ be a spherical fusion category, and put $d = \dim \mathcal{S}$. For $A \in \operatorname{Irr} \mathcal{Z}(\mathcal{S})$, let $a_A = \dim A$ and θ_A be its twist. Then a_A is real and*

$$\sum_A a_A^2 = d^2, \qquad \sum_A a_A^2 \theta_A = d, \qquad \sum_A a_A^2 \theta_A^2 = d \sum_V \dim(V) \nu_2(V).$$

In particular the coefficients a_A^2 form a positive measure on roots of unity, even when some dimensions of $\mathcal{S}$ are negative.

Proof. Simple dimensions in a spherical fusion category are totally real [13, Section 2.1]. The forgetful functor is spherical, so a_A is an integral linear combination of these real dimensions. Let I be central induction and $m_{V,A} = [I(V) : A]$. Adjunction gives $a_A = \sum_V m_{V,A} \dim V$. The indicator identity of [11, Theorem 4.1]

$$\operatorname{Tr}(\theta^n|_{I(V)}) = d\nu_n(V)$$

therefore gives, after multiplication by $\dim V$ and summation over V, $\sum_A a_A^2 \theta_A^n = d \sum_V \dim(V) \nu_n(V)$. For $n = 1$ only the tensor unit contributes, and for $n = 2$ this is the claimed formula. The zeroth moment is the standard identity $\dim \mathcal{Z}(\mathcal{S}) = d^2$. The twist is a root of unity by the Vafa theorem for the center of a spherical fusion category. □

Proposition 6.2 (A global codegree moment bound)**.** *Let a commutative fusion ring have a spherical categorification, with simple dimensions d_i, global dimension $\Delta = \sum_i d_i^2$ and formal codegrees f_j, counted with multiplicity. Put*

$$w = \sum_j f_j^{-2}, \qquad S = \sum_{i=i^*} |d_i|, \qquad h = \Delta^2 w.$$

At every real embedding of its dimension field one has

$$2(\Delta - h)^2 \le (\Delta^2 - h)(\Delta^2 + \Delta S - 2h). \tag{16}$$

Equivalently, $\Delta(1-3w) + (1-w)S + 4w - 2/\Delta \ge 0$. *For a self-dual pseudo-unitary category,* $S = \sum_i \mathrm{FPdim}(X_i)$.

Proof. The constituents of $I(1)$ are distinct, have twist one and dimensions Δ/f_j. Remove their squared mass h from the positive measure of Lemma 6.1. The remaining mass is $\Delta^2 - h$, its first moment is $\Delta - h$, and its second moment is $\Delta s - h$, where $s = \sum_i d_i \nu_2(X_i) \le S$. The weighted sum of squared real parts of twists is $(\Delta^2 + \Delta s - 2h)/2$. Cauchy–Schwarz gives the displayed inequality with s in place of S; increasing s to S preserves it because the remaining mass is nonnegative. Expanding and dividing by $\Delta^3 > 0$ gives the equivalent expression. The argument applies to every Galois conjugate spherical structure. The rational number w can be computed without finding codegree roots: $w = \mathrm{Tr}(B^{-2})$ for the Casimir matrix B. □

This form of the bound applies to every possible spherical dimension character, including those outside the Frobenius–Perron orbit. In the finite certificates it excludes the remaining character branches for $F_{21,325}$ and $F_{24,735}$; a pseudo-unitary obstruction alone would not exclude those branches.

Proposition 6.3 (The signed-translate moment obstruction)**.** *Let* $\mathcal{S}$ *be a sphericalization with* $\dim \mathcal{S} = 2\delta$ *and commutative Grothendieck ring. Write* A_j *for the simple constituents of* $I(1)$ *and* $a_j = \dim A_j > 0$. *Put*

$$h = \sum_j a_j^2, \qquad 2s = \sum_{V \in \mathrm{Irr}\,\mathcal{S}} \dim(V)\nu_2(V).$$

Then there are signs $\eta_j \in \{1, -1\}$, *with* $\eta_j = -1$ *for* $A_j = 1$, *such that*

$$\sum_j \eta_j a_j = 0.$$

For $C = \sum_j \eta_j a_j^2$, *one necessarily has*

$$h \le 2\delta^2, \qquad h/\delta - \delta \le s \le \delta,$$
$$(2\delta - h - C)^2 \le (4\delta^2 - 2h)(2\delta s + 2\delta^2 - 2h).$$

These conditions hold after every Galois conjugation, with the same second indicators and twist signs.

Proof. The formal-codegree theorem [13, Theorem 2.13] shows that $I(1)$ is multiplicity free and its constituents have positive dimensions; their twists are one by [13, Theorem 2.5]. The projection formula gives $I(\varepsilon) = \varepsilon \otimes I(1)$, whose constituents have dimensions $-a_j$ and are therefore disjoint from those of $I(1)$. Their twists η_j are ± 1, by balancing and the fact that the monodromy with an invertible object of order two squares to one. The indicator identity for $I(\varepsilon)$ gives $\sum_j \eta_j a_j = 0$; the constituent $\varepsilon \otimes 1$ has twist -1.

Remove these two sets of constituents from the positive measure of Lemma 6.1. The remaining mass is $M = 4\delta^2 - 2h$, its first moment is $2\delta - h - C$, and its second moment is $4\delta s - 2h$. If its twists are $e^{i\phi_A}$, then

$$\sum_{\text{remaining}} a_A^2 \cos^2 \phi_A = 2\delta s + 2\delta^2 - 2h.$$

The inequalities $0 \le \sum a_A^2 \cos^2 \phi_A \le M$ give the two bounds on s. Cauchy–Schwarz gives the last inequality. Every dimension remains real under the Galois embeddings under consideration, whereas the second indicators and the twist signs are rational numbers and remain fixed. □

Corollary 6.4 (Finite signed-moment certificate)**.** *Let T be a proposed signed table from Proposition 2.5, and let χ be a proposed dimension character, with $\delta = \sum_i \chi(b_i)^2$. Form the eight numbers $a_j = \delta/f_j$, where the f_j are the formal codegrees of R and T, with all multiplicities. Choose the occurrence $f_j = \delta$ corresponding to the central unit. For every self-dual nonunit basis element choose $\nu_i \in \{-1, 1\}$ and put*

$$s = 1 + \sum_{i>0,\ i=i^*} \nu_i \chi(b_i).$$

If no joint choice of these indicators and the twist signs η_j satisfies Proposition 6.3 at every real embedding, T cannot be the signed quotient of a sphericalization with dimension character χ.

Proof. The two lifts of a non-self-dual object have zero second indicator. For a self-dual object the two lift dimensions and indicators both change sign, so their products agree. The unit and its sign translate contribute two. This gives the displayed s and exhausts the indicator possibilities. The formal-codegree description in Proposition 2.5 gives the eight a_j. The remaining assertions are exactly the necessary conditions of Proposition 6.3. Equal codegrees retain separate occurrences; an occurrence of dimension one may be designated the unit without losing any possibility, since the other signs are all enumerated. □

The supplied certificate for $F_{17,308}$ uses this corollary in one character branch. The other signed branches are handled by the same arithmetic, spherical and parity routines as all other rings.

7 An adjoint-parity obstruction

The following elementary property of pivotal duality gives an additional constraint on the signed tables. It does not require a braiding, pseudo-unitarity or a choice of dimensions.

Lemma 7.1 (Skew simples in adjoint products)**.** *Let $\mathcal{C}$ be a pivotal fusion category over $\mathbb{C}$, and let X be a self-dual simple with $\nu_2(X) = -1$. For every object Y, the multiplicity of X in $Y^* \otimes Y$ is even.*

Proof. Write $j_V : V \longrightarrow V^{**}$ for the pivotal structure. On $\mathrm{Hom}(V, V^*)$ use the transpose $f^t = f^* j_V$. The definition of the second indicator gives an isomorphism $f : X \longrightarrow X^*$ with $f^t = -f$. Put $W = Y \otimes X \otimes Y^*$ and

$$\alpha = j_Y \otimes f \otimes \mathrm{id}_{Y^*} : W \longrightarrow W^*.$$

Duality reverses tensor factors. The identity $j_Y^* j_{Y^*} = \mathrm{id}_{Y^*}$ and monoidality of j therefore give $\alpha^t = -\alpha$. Associativity and duality coherence maps are understood in this formula.

On the complex vector space $\mathrm{Hom}(1, W)$ the pairing

$$B(v, w) = v^* \alpha w$$

is nondegenerate: duality identifies this space with $\mathrm{Hom}(W^*, 1)$, composition with α is an isomorphism, and the composition pairing on the unit isotypic component is perfect by semisimplicity. Dualizing the scalar $B(v, w)$ and using $v^{**} = j_W v$ gives $B(v, w) = w^* \alpha^t v = -B(w, v)$. Thus its dimension is even. Rigidity identifies its dimension with $\dim \mathrm{Hom}(X, Y^* \otimes Y)$, proving the assertion. □

Corollary 7.2 (A congruence for sphericalization)**.** *Let R, T be an unsigned and signed pair as in Proposition 2.5, arising from a categorification. For every self-dual basis element b_i there is a sign $\nu_i \in \{1, -1\}$ such that, for every j,*

$$N^i_{j,j^*} - \nu_i T^i_{j,j^*} \equiv 0 \pmod 4. \tag{17}$$

Consequently, if neither sign satisfies these congruences for some i, the signed table cannot arise from a sphericalization.

Proof. The two lifts X_i^+, X_i^- have opposite second indicators, as explained in Section 6. Take $\nu_i = \nu_2(X_i^+)$. In the adjoint product of a chosen lift of b_j, their multiplicities are respectively $(N^i_{j,j^*} + T^i_{j,j^*})/2$ and $(N^i_{j,j^*} - T^i_{j,j^*})/2$. The lift whose indicator is -1 has even multiplicity by Lemma 7.1; this is precisely (17). The same ν_i must work for all j. □

The verifier applies this congruence to every self-dual basis element of every retained signed table. A witness records an index for which both indicator choices fail, and the checker recomputes the unsigned and signed adjoint coefficients from the two tensors. Thus a parity exclusion depends only on the fusion data. Its occurrences, including the signed branches it resolves, are tabulated automatically in the certificate ledger.

7.1 Joint use of arithmetic, parity and moments

The preceding criteria constrain the same hypothetical sphericalization. They may therefore be imposed jointly, before expanding its possible signed tables or indicator assignments.

Corollary 7.3 (Joint necessary conditions for a signed lift)**.** *Let R, T, χ be as in Proposition 2.5, and put $K = \mathbb{Q}(\chi(b_i) : i)$. If these data arise from a sphericalization, then both of the following conditions hold.*

1. *Every formal codegree of T is a d-number, and every value of every character of T is a cyclotomic integer.*
2. *The indicators in Corollary 6.4 satisfy*

$$N^i_{j,j^*} - \nu_i T^i_{j,j^*} \equiv 0 \pmod 4 \qquad \text{for every } j$$

for each self-dual nonunit b_i. One assignment of these indicators and the twist signs η_j must satisfy the signed-translate moment inequalities at every real embedding of K.

Proof. The characters of T are the characters of the sphericalization's Grothendieck ring on which the central sign object acts as -1. Its formal codegrees on this summand are $2f$, with f a formal codegree of T. Proposition 2.1 applies: multiplication by a nonzero rational number preserves the d-number property, and the character values remain cyclotomic integers.

The congruences are Corollary 7.2 and concern the same second indicators as the moment identities. Their choices must therefore agree. Under Galois conjugation the indicators and relevant twist signs remain fixed, since they lie in $\{1, -1\}$. The dimensions remain real and the squared-dimensional measure remains positive, so Proposition 6.3 applies at every real embedding with that same assignment. □

Corollary 7.4 (Parity-refined spherical bound)**.** *In Proposition 6.2, replace S by*

$$S_{\mathrm{par}} = 1 + \sum_{i \in J} d_i + \sum_{\substack{i>0,\ i=i^* \\ i \notin J}} |d_i|, \qquad J = \{i > 0 : i = i^*,\ N^i_{j,j^*} \text{ is odd for some } j\}.$$

The resulting inequality remains necessary at every real embedding.

Proof. Lemma 7.1 forces $\nu_2(X_i) = 1$ for $i \in J$. Thus $s = \sum_i d_i \nu_2(X_i) \le S_{\mathrm{par}}$ at every real embedding. The proof of Proposition 6.2 uses this upper bound only after multiplication by the nonnegative remaining mass $\Delta^2 - h$, so the replacement preserves the necessary inequality. □

Corollary 7.5 (Parity-restricted signed enumeration)**.** *For a self-dual rank-four fusion ring, write the unsigned and signed parameter tuples as $N = (N_0, \dots, N_9)$ and $p = (p_0, \dots, p_9)$ in the order $(a, b, c, d, e, f, g, h, i, j)$. Put*

$$J_1 = \{0, 3, 5\}, \qquad J_2 = \{1, 6, 8\}, \qquad J_3 = \{2, 7, 9\}.$$

It suffices to search the union, over $\nu = (\nu_1, \nu_2, \nu_3) \in \{1, -1\}^3$, of the domains

$$|p_k| \le N_k, \qquad p_k \equiv \nu_i N_k \pmod 4 \quad (k \in J_i), \qquad p_4 \equiv N_4 \pmod 2.$$

One may intersect these domains with the prescribed odd-coefficient positivity normalization. After duplicate and empty domains are removed, the resulting at most eight boxes are pairwise disjoint.

Proof. Corollary 7.2 supplies three indicator signs for every categorical signed table. The sets J_i are precisely the coefficients of the three nonunit simples in the three adjoint products. The remaining parameter e retains its original signed parity and size restrictions.

For fixed i, the domains for $\nu_i = 1$ and $\nu_i = -1$ coincide if every N_k, $k \in J_i$, is even. Otherwise an odd N_k gives distinct residues modulo four, making these domains disjoint. Intersecting with positivity restrictions preserves this dichotomy, except that a domain may become empty. Since the sets J_i are disjoint, the assertion about the boxes follows.

A basis-sign change $s_i \in \{1, -1\}$ multiplies $T^i_{j,j}$ by s_i and changes the chosen lift's indicator to $s_i \nu_i$. The product $\nu_i T^i_{j,j}$ is unchanged. Thus the union is preserved by basis-sign changes, and the positivity normalization retains a representative of every categorical possibility. □

The omitted coefficient tuples violate adjoint parity, whether or not they are associative. A search restricted to these domains therefore counts parity-admissible normalized associative tables. Coincident domains do not identify their indicator choices: all compatible indicators remain available in the joint moment test. These refinements use the same categorical criteria; they change their order of application and the size of the finite searches.

The arithmetic conditions can also be applied once to a common bounded set of signed tables before testing compatibility with any unsigned ring. Appendix C proves completeness of such a shared cover under basis permutations and sign changes. Every compatible transform of a surviving orbit is still checked against the unsigned ring's bounds, parities and remaining moment conditions.

8 Exact certification and completion

8.1 From theorems to a verifier

A certificate is a finite witness to one of the preceding necessary conditions. It is not an unexplained exclusion flag. The basic records are an irreducible factor and failed divisibility, a prohibited based subring, a fully checked family recognition, or a finite-field factorization. A signed record identifies its exhaustive search or shared-cover method, contains every table remaining after its verified necessary prefilters, and gives a proof for every remaining character component. An induction record contains all roots of the codegree polynomial in the dimension field and all admissible low restriction choices, with a violated Gram or moment condition for each.

For character computations, take $H = 1 + b_1 + kb_2 + k^2 b_3$. A separating choice exists for $k \in \{1, \dots, 13\}$: for each of the six pairs of distinct characters, equality on H excludes at most two values of k. The squarefree characteristic polynomial of N_H then identifies the rational algebra with $\mathbb{Q}[t]/(\mathrm{charpoly}(N_H))$. The checker expresses every N_i as a polynomial in N_H and verifies these identities exactly. Factoring the defining polynomial enumerates all character-field components. The strictly positive matrix N_H identifies the Frobenius–Perron component by its largest real eigenvalue, isolated over $\mathbb{Q}$.

There are two different finite completeness checks. The signed search must cover every bounded coefficient choice modulo its stated gauge; the low-induction search must cover every vector in (8) and every allowable column combination. Both are regenerated during verification. In particular a supplied list of solutions is never presumed exhaustive. Polynomial roots are checked by multiplying the linear factors back to the original polynomial. Signs of algebraic numbers are decided by rational isolating intervals and refinement, not by floating-point tolerances. Details of the arithmetic implementation and its elimination branches are recorded in the verification guide.

Proposition 8.1 (Soundness of the certificate scheme)**.** *If the verifier accepts a nonexistence certificate for a fusion ring R, then R has no complex fusion categorification, assuming the established inputs in Proposition 2.1 and the indicator and central induction theorems cited in the proofs.*

Proof. The direct certificates apply those inputs, the scalar arithmetic lemmas, Theorem 3.4, Proposition 5.6, or the uniform applications in Section 5. A weakly integral or two-dimension record first reduces general existence to pseudo-unitarity, with Galois conjugation when required. A pseudo-unitary record applies Proposition 3.3 or the root-length obstruction, or the global codegree moment bound. A spherical record exhausts all character components, rejecting zero values, unequal values on duals, a failed norm or field condition, a Galois orbit with a verified pseudo-unitary obstruction, or Proposition 6.2 at a real embedding.

For a general signed record, Proposition 2.5 supplies at least one table in the exhaustive bounded search. For a shared search, Proposition C.1 supplies its global orbit; the checker verifies that cover before accepting any record depending on it and tests every compatible transform not already excluded by a common arithmetic obstruction. A rejected Casimir factor cannot be its global dimension by Lemma 2.3; the prime determinant shortcut is a consequence of the same argument. The same established d-number and cyclotomicity conditions also apply to the signed quotient, by the joint necessary conditions proved above. An unsigned gauge yields a spherical categorification, excluded by its separate spherical record. Otherwise every possible dimension-character component is rejected by a norm or field certificate or by Corollary 6.4; a whole signed table may instead be rejected by Corollary 7.2. Parity is imposed on the same indicator assignments used in the moment identities, and its coefficient-domain restriction preserves every categorical possibility. Thus the hypothetical category has no remaining branch. These implications have no cyclic dependence on the classification conclusion or on the companion paper. □

8.2 The exhaustive run

Proof of Theorem 1.1. The complete pinned census [1] gives 889 530 pairwise nonisomorphic based rings of multiplicity at most 128. The checker reconstructs every full multiplication tensor and checks the fusion axioms. For each of the fifteen positive entries, the verifier checks a based-ring isomorphism to the full tensor of a cited model in Table 1. Those established realizations prove existence.

For every other entry the accepted certificate has one of the types in Table 3. The counts sum to 889 515. The 7 084 signed records collectively enumerate 801 047 normalized associative

candidates after their stated necessary prefilters, with every zero-denominator branch retained. Candidates omitted before enumeration are eliminated by the stated necessary filters; every enumerated candidate has a checked rejection. In particular, neither a ring identifier nor a preassigned classification status is accepted as a mathematical witness. Proposition 8.1 therefore gives general nonexistence for all remaining rings. Unitary existence for the positive rings and general nonexistence for all others identify the five existence classifications. □

First successful top-level test, in dispatch order	First use m	Rings
Formal-codegree d-number failure	1	872 625
Rank-two moment obstruction	2	1 273
Non-self-dual quadratic moments	2	127
Diagonal involution moments	3	126
Residual-involution propagation	5	1
C_3 near-group root length	9	37
Automatic near-group classification	18	3
Cyclotomicity obstruction	9	84
Fusion-subring dimension divisibility	2	28
Weak integrality and induction	2	1
Two-dimension Galois reduction and induction	3	8 126
Signed sphericalization	2	7 084
Total excluded		889 515

Table 3: The actual order of the first successful top-level tests. A model is recognized before these tests. First use is the least exact multiplicity with a certificate of this type; nested tests in signed records are not counted again. The sequence is chosen for high early rejection and short recognition before costly signed searches.

The finer ledger groups individual checks by their underlying mathematical criterion. Its latest first use is at multiplicity 18. In this fixed certificate allocation, all multiplicities from 19 to 128 require no additional criterion family. This describes the completed run, not a proof that the collection of criteria is minimal.

The numerical tables in this paper are generated from accepted certificates and the pinned census. The complete per-ring ledger, per-multiplicity counts, witness files, and criterion-use statistics are supplied electronically. No manually calculated census case is needed for the proof. The categorical arguments precede their finite application; the code only recognizes their hypotheses and checks their exact algebraic consequences.

8.3 Compact certificates and reproducibility

The supplement shares a complete bounded signed census across the unsigned rings. Proposition C.1 proves that enumeration modulo all basis-sign changes and nonunit basis permutations covers every possible signed quotient. Universally invalid signed orbits are rejected once by formal-codegree d-numbers or cyclotomicity. For each unsigned ring, the verifier expands every remaining orbit into all compatible transforms and imposes the signed coefficient bounds, parities, adjoint congruences, and determinant-prime conditions. Every remaining table receives its own character or induction certificate. No choice of a convenient orientation substitutes for this full expansion.

The global cover and all its common arithmetic rejections are recomputed from source during verification; a stored list is not presumed exhaustive. The smaller direct backends

use complete per-ring enumeration, optionally restricting coefficient domains by adjoint parity first. Every record specifies its method. In all methods the checker reconstructs the candidate set, compares it with the complete supplied list, and verifies each longer certificate. A missing, extra, or altered candidate causes rejection. The reported candidate count sums the explicitly visited tables after each method's stated prefilters; it does not count every raw signed coefficient choice, or count universally excluded orbits repeatedly. The verified common cover contains 6 470 006 signed orbits, of which 30 207 survive the universal arithmetic tests. The global orbit counts and the method-specific totals are recorded separately in the verification guide and electronic reports. An independent arbitrary-integer Python implementation supports comparison with the direct bounded searches.

Formal-codegree arithmetic is accelerated by exact integer matrix and polynomial operations. The verifier reconstructs the Casimir polynomial, checks its irreducible factors and the failed divisibilities, and uses rational isolating intervals for ordered algebraic comparisons. Floating point approximations never decide a categorification status. Source hashes, dependency versions, the pinned census commit, and the commands for a complete replay are recorded in the verification guide. The verification of a finite certificate and the census completeness result documented in [1] are separate inputs, and both are required.

Conjecture 8.2 (Unrestricted classification). The fifteen rings in Table 1 are all the rank-four Grothendieck rings of complex fusion categories, without any multiplicity restriction.

Conjecture 8.3 (Sufficiency of the certificate criteria). Every rank-four fusion ring outside Table 1 has a finite nonexistence certificate using the necessary criteria and signed reduction of this paper.

The second conjecture is stronger: it predicts that no additional categorical obstruction will be needed, with arbitrary exact witnesses allowed for the stated arithmetic tests. It concerns the mathematical criteria, not fixed prime-search limits or machine-integer bounds in an implementation. Its evidence is the complete certificate coverage of 889 515 exclusions through 128, together with the unrestricted families handled by the uniform arguments. The two conjectures assert no theorem above this bound. A uniform obstruction or bound for the remaining families is still required. The companion paper develops unrestricted reductions and isolates the unresolved cubic and quartic pivotal cases.

The supplementary command `python verify.py` performs the complete bounded verification. Its default signed-table backend is the supplied C++17 enumerator; an arbitrary-integer Python backend is provided for independent comparison. The mathematical proof does not require numerical eigenvalues, pentagon solving or modular-data classification. The computation certifies the finite algebraic applications; it is not a proof-assistant formalization of the categorical arguments.

A Arbitrary twist sectors

For $\zeta \in \mathbb{C}^\times$, let $I(V)_\zeta$ denote the part of $I(V)$ with twist ζ, including multiplicities. This appendix is not required for the finite classification; its twist-one instance is already used in Theorem 3.1.

Proposition A.1 (Twist-sector bound). *For every object V in a pseudo-unitary fusion category,*

$$2\,\mathrm{FPdim}(I(V)_\zeta) \le D\big(\mathrm{FPdim}(V) + [V:1]\big).$$

Proof. Put $n = [V:1]$ and $W = \mathrm{FPdim}(I(V)_\zeta)$. If $W = 0$ there is nothing to prove. Otherwise $|\zeta| = 1$, and the first indicator gives $nD = \zeta W + \sum_{\theta_A \neq \zeta} w_A \theta_A$. The triangle inequality implies $W \le nD + D\,\mathrm{FPdim}(V) - W$. □

Corollary A.2 (Higher-indicator twist sectors)**.** *If $\nu_1(V) = \cdots = \nu_m(V) = 0$, then, for every $\zeta \in \mathbb{C}^\times$,*

$$(m+1)\,\mathrm{FPdim}(I(V)_\zeta) \le D\,\mathrm{FPdim}(V).$$

Proof. For a nonzero sector $|\zeta| = 1$. Apply the proof of Theorem 3.1 to $|1 + \zeta^{-1}\theta_A + \cdots + (\zeta^{-1}\theta_A)^m|^2$. The vanished moments remain zero after these rotations. □

B Census input and tensor conventions

The classification takes its exhaustive list of rank-four based rings from [1], pinned to the commit specified in Section 1. The repository gives its enumeration method, canonicalization, counts, and full tensor data. We recall the coefficient conventions needed to reconstruct the local certificates and the signed search. All parameters below are integers between 0 and the multiplicity bound M.

B.1 Self-dual basis

Frobenius reciprocity gives the regular matrices

$$N_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & c \\ 0 & b & d & e \\ 0 & c & e & f \end{pmatrix}, \quad N_y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & d & e \\ 1 & d & g & h \\ 0 & e & h & i \end{pmatrix}, \quad N_z = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & c & e & f \\ 0 & e & h & i \\ 1 & f & i & j \end{pmatrix}.$$

Associativity is equivalent to their pairwise commutativity, or to

$$ad - b^2 + bg + ch - d^2 - e^2 + 1 = 0, \tag{18}$$
$$ae - bc + bh + ci - de - ef = 0, \tag{19}$$
$$be - cd + dh - eg + ei - fh = 0, \tag{20}$$
$$af + bi - c^2 + cj - e^2 - f^2 + 1 = 0, \tag{21}$$
$$bf - ce + di - eh + ej - fi = 0, \tag{22}$$
$$df - e^2 + gi - h^2 + hj - i^2 + 1 = 0. \tag{23}$$

The equivalence follows by comparing the first row of products of regular matrices and then using commutativity to recover every other row. It is also checked directly against the full associativity identities. For the signed-cover Diophantine reduction, when $b > 0$ define

$$H_0 = e(d+f) + bc, \qquad G_0 = b^2 + d^2 + e^2 - 1,$$
$$A = e(b^2 - c^2) + bc(f - d), \qquad C = e(bf - ce),$$
$$B_0 = b^3 e - b^2 cd + b(d-f)H_0 - ebG_0 + ecH_0.$$

Eliminating h, g, j from the six equations yields the necessary identity

$$Ca + Ai + B_0 = 0. \tag{24}$$

The signed verifier checks all six original equations after using this identity, retaining every vanishing-coefficient branch.

B.2 One nontrivial dual pair

For $X^* = X$ and $Y^* = Z$, the regular matrices are

$$N_X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & b \\ 0 & b & c & d \\ 0 & b & d & c \end{pmatrix}, \quad N_Y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & c & d \\ 0 & d & e & f \\ 1 & c & e & e \end{pmatrix}, \quad N_Z = N_Y^{\mathsf{T}}.$$

Associativity is equivalent to

$$\begin{aligned} ac - b^2 + 2be - c^2 - d^2 + 1 &= 0, & ad - b^2 + be + bf - 2cd &= 0, \\ b(c-d) + d(e-f) &= 0, & -c^2 + d^2 - e^2 + f^2 - 1 &= 0. \end{aligned}$$

The pinned source contains 889530 classes through $M = 128$; `data/parameters_128.txt.gz` is its compact snapshot and `data/census_128_manifest.json` records the source commit and digests. The verifier reconstructs each full tensor, checks its axioms and digest, and tests its categorification certificate. The exhaustive census method and its count certification are supplied by the source repository [1]; no second rank-four census is required by the classification.

C A shared exhaustive signed cover

The signed reduction permits an arithmetic exclusion to be shared among all unsigned rings compatible with a signed table. The following construction makes that shared search exhaustive.

Proposition C.1 (A shared exhaustive signed cover)**.** *Fix $M \geq 1$. Consider the integral self-dual rank-four signed tables with the prescribed unit and reciprocity, parameter tuple $(a, b, c, d, e, f, g, h, i, j)$, and all ten coefficients of absolute value at most M. The six associativity equations define a finite set preserved by the group of nonunit basis permutations and basis-sign changes, of order* 48*.*

Every orbit has a representative satisfying

$$b, c, d \geq 0, \qquad b = \min\{|b|, |c|, |d|, |f|, |h|, |i|\}, \qquad e \geq 0 \text{ if } b = c = 0.$$

Enumerating all such representatives and retaining only the lexicographically least preferred tuple in each of their 48 *transforms gives exactly one table per orbit.*

For every self-dual unsigned fusion ring R of multiplicity at most M, all its possible sphericalization quotients are covered by the transforms of these representatives that satisfy its signed coefficient bounds, parities, and adjoint-parity conditions. A representative failing the formal-codegree d-number criterion or character cyclotomicity can be discarded for every R simultaneously.

Proof. The permutation group is transitive on the six mixed positions b, c, d, f, h, i. Place a position of minimum absolute value at b. Under a basis-sign change (s_1, s_2, s_3), the signs multiplying b, c, d are respectively s_2, s_3, s_1. Hence these three entries can be made nonnegative independently, without changing their absolute values. If $b = c = 0$, the remaining choices of s_2, s_3 also make e nonnegative while preserving $d \geq 0$. Thus the preferred set meets every orbit. Its least element in each finite orbit is unique; checking all 48 transforms selects precisely that element.

For completeness, the preferred set is exhausted by the following disjoint branches of the associativity equations. If $b > 0$, then $c, d \geq b$ and $|f|, |h|, |i| \geq b$; the two-coefficient

Diophantine relation (24), with the polynomial coefficients from Appendix B, holds for signed parameters as well. Exhaust the bounded choices of b, c, d, e, f and solve that relation for all bounded integer pairs a, i by exact greatest-common-divisor progressions. If one coefficient vanishes, exhaust its free variable; if both vanish, retain the entire bounded product precisely when the constant term vanishes. The formulas in that appendix then determine h, g, j. Impose the signed bounds and the preferred-representative conditions on all coordinates, and check the original six equations. If $b = 0 < c$ and $e \neq 0$, the equations determine

$$h = \frac{d^2 + e^2 - ad - 1}{c}, \qquad i = \frac{e(d + f - a)}{c},$$
$$j = \frac{c^2 + e^2 + f^2 - af - 1}{c}, \qquad g = \frac{h(d - f) + ei - cd}{e}.$$

The divisibility for i gives $a \equiv d + f \pmod{c/\gcd(c, e)}$; all bounded choices in this progression are checked. If $b = 0 < c$, $e = 0$ and $d > 0$, then

$$\gcd(c, d) = 1, \quad i = 0, \quad h = \frac{d^2 - ad - 1}{c} \neq 0, \quad f = d - \frac{cd}{h},$$

and j is given by the preceding formula, while g is enumerated. The remaining case $b = 0 < c$, $e = d = 0$ forces $(c, h, f, i, j) = (1, -1, 0, 0, 0)$ and leaves a, g free.

If $b = c = 0 < e$, then $a = d + f$ and $df = e^2 - 1$; the equations determine g, j from d, e, f, h, i, with the case $d = 0$ retained. Finally, $b = c = e = 0$ forces $d = 1, a = 0$ and $f \in \{1, -1\}$. For $f = 1$ the remaining equation is

$$gi + hj = h^2 + i^2 - 2,$$

including its $h = 0$ branch. For $f = -1$ one has $h = i = 0$ and g, j are free. Coordinates whose signs are not fixed by the preferred convention retain both signs and the original size bounds, and every emitted tuple is checked against all six equations. This proves exhaustive enumeration of the preferred set, including the zero denominators.

Any sphericalization quotient for R lies in the bounded signed set by Proposition 2.5. One of the 48 transforms of its global representative is the quotient in the basis of R, so testing all compatible transforms loses no categorical possibility. Formal codegrees are unchanged by permutations and basis signs, and character values change only by permutation and sign. Their d-number and cyclotomicity conditions are therefore orbit-invariant. Both conditions are necessary for a signed sphericalization quotient by Corollary 7.3, which proves the common exclusions. □

Remark C.2 (Certificate reuse). The global signed census is regenerated exhaustively during the verification workflow; its completeness is not inferred from a cached list. Universally excluded orbits receive a checked common arithmetic rejection. Every other orbit is expanded into all distinct transforms compatible with each unsigned ring, and every such candidate receives the corresponding signed-branch check. Sharing these computations changes their allocation among records, not the necessary criteria or the scope of the classification.

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GPT-6 Astra assisted with research, computation, and manuscript preparation.

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